ENGL 220 | SPRING 2026 | DR. NAQAA ABBAS

WALKABILITY IN EDUCATION CITY

Addressing Pedestrian Comfort Through Sustainable Solutions

Team Cocktail

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THE PROBLEM: WALKING IN THE HEAT

40°C+
Summer temperatures

57%
Drive private vehicles

17%
Walk between facilities

Why It Matters

- EC spans 12 km² with 24 tram stations, yet extreme heat and unshaded walkways make walking unbearable.
- Increased carbon emissions, reduced physical activity.
- Students, faculty members, staff, and visitors are most affected.
- Contradicts with QNV 2030 targets

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QNV 2030 targets: 25% emissions reduction • 60–70% pedestrian shade • Sustainable transport

SOLUTIONS EVALUATED (1 OF 2)

1. Park and Plant Initiative

Native Sidra & Ghaf trees along walkways to create shaded corridors

- 75% canopy shading optimizes airflow & cooling
- Started with 2,500 trees → target 20,000
- Dubai study: warm/hot hours dropped from 1,291 to <300
- Community programs: 'Adopt a Tree', 'Green Lungs'

Limitation: 3-5 years to full canopy maturity

2. Cool Pavement Coatings

Reflective layers on walkways to reduce heat absorption

- Surface temps reduced by 2.6–4.9°C
- Low cost, minimal construction required
- Easy campus-wide application

Limitation: May increase radiant temperature at pedestrian level by 2–3°C, making walkers feel hotter. Degrades with UV/dust.

SOLUTIONS EVALUATED (2 OF 2)

3. Enclosed Walkways

Air-conditioned enclosed corridors for full climate control

- Best thermal comfort of all solutions
- Qatar precedents: Industrial Area bridge, Mall of Qatar connection
- Protects vulnerable groups (children, elderly)

Limitation: Massive energy demands contradict sustainability goals. High cost, impractical for EC's spread-out layout. Reduces open social spaces.

4. Solar-Powered Misting

High-pressure misting nozzles powered by solar panels

- Shade + misting reduces thermal index by 12.3°C
- Proven at Expo 2020 Dubai & Saudi Arabia
- Smart sensors activate only when needed

Limitation: Effectiveness drops at humidity >60-70%. Qatar's summer humidity reaches 40-60%. Limited to existing canopy areas. Nozzle maintenance issues.

WEIGHTED DECISION MATRIX

Criteria	Weight	Park & Plant	Cool Pavement	Enclosed Walkways	Solar Misting	EC Canopies (Reference)
Thermal Comfort	30%	**3 (0.90)**	1 (0.30)	5 (1.50)	4 (1.20)	3 (0.90)
Sustainability	30%	**5 (1.50)**	2 (0.60)	1 (0.30)	4 (1.20)	3 (0.90)
Coverage & Scalability	25%	**4 (1.00)**	5 (1.25)	2 (0.50)	3 (0.75)	3 (0.75)
Feasibility	15%	**4 (0.60)**	4 (0.60)	1 (0.15)	3 (0.45)	3 (0.45)
TOTAL SCORE		**4.00**	2.75	2.45	3.60	3.00

Park and Plant scores highest (4.00/5.0) — the most balanced and context-appropriate solution

RECOMMENDATION: EXPAND PARK AND PLANT

Already Operational

Existing QF program with staffing, budgets, and 2,500+ trees planted

Eco-Friendly

Sequesters carbon, cleans air, improves biodiversity

Compounding Impact

Shade and cooling grow stronger every year as trees mature

Climate-Proof

Effective in any humidity — unlike misting systems

Implementation Phases

1

Phase 1 (Years 1–2)

High-traffic corridors linking tram stations to key buildings

2

Phase 2 (Years 2–4)

Expand to housing, parking areas, and Oxygen Park

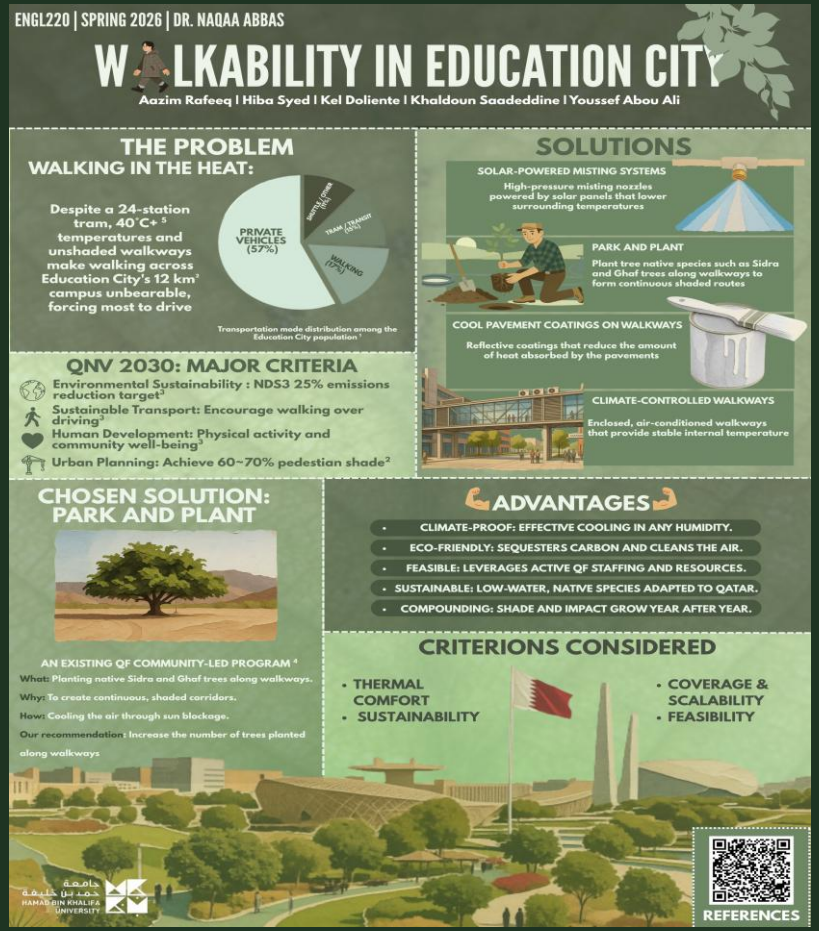
3

Phase 3 (Years 4–6)

Achieve 60–70% campus-wide shade coverage

Aligns with QNV 2030: Environmental sustainability, human development, and 60–70% shade target

OUR RESEARCH POSTER



THANK YOU

Questions?

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